Maximizing Revenue and Minimizing Risk in the Hotel Business

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Abstract

The hotel industry operates in an environment of high uncertainty, with fluctuating demand, and complex customer behavior. A key challenge for hotel operators is managing revenue volatility and operational risks, particularly cancellation rates. Our data shows that more than one in three hotel bookings is canceled, creating significant financial and planning uncertainty. If not managed effectively, cancellations lead to revenue loss, inefficient room allocation, and suboptimal pricing decisions. To maximize revenue while minimizing risk, hotels need data-driven forecasting strategies. This paper's research question is that how machine learning can help us identify business strategies for cancellation patterns, and booking lead time.

1 Introduction

Booking cancellations present ongoing challenges for hotels, impacting both revenue and operational planning. Effectively managing cancellations requires a clear understanding of why they happen and its accurate prediction as shown in Urrea et al. (2024). By integrating descriptive analytics and clustering methods this paper provides practical insights to help hotels move from reactive approaches toward proactive revenue management.

2 Data Description

This paper utilizes the publicly available *Hotel Booking Dataset* from Kaggle (Kaggle, 2024), with 116,958 hotel reservation records between 2015 and 2017.

The dataset includes 34 variables whereas relevant for this paper are **Lead Time** (days between booking and arrival), **Average Daily Rate** (ADR), and **Cancellation Status** (binary whether booking was canceled). We pre-processed the dataset by removing two

columns namely 'company', and 'agent' (which had the majority of entries with missing values), dropping the rest of rows with missing values, and filtering out invalid ADR values, resulting in a cleaned dataset with 2,432 fewer rows (2.04% decrease).

3 Methods

3.1 Fundamental Analysis

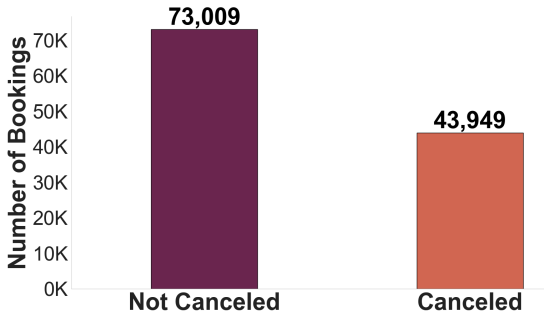

Figure 1: Total Number of Canceled and Non-Canceled Bookings.

The overall cancellation rate, as shown in Figure 1, reveals that out of all 116,958 bookings, 43,949 (approximately 38%) were canceled. This highlights the significant impact and risk of cancellations on hotel revenue and operations.

To understand the value of cancellations, we analyzed a correlation heatmap A5 and found that high-ADR bookings tend to cancel more often if booked short-term and long lead-time bookings show a higher overall cancellation rate, making them a major factor in long-term revenue planning. Additionally, bookings for higher-priced rooms made far in advance are less likely to be retained as lower-prices rooms as seen in the Survival Analysis A7. Our Analysis of Figure A6 found that the business should pay attention on cancellations for peak months in January and July.

3.2 Clustering with Machine Learning

To analyze customer clusters we used three algorithms:

K-Means is a centroid-based method that efficiently groups data points by assigning them to the nearest cluster center, making it a practical choice for predefined similarity metrics.

Gaussian mixture model (GMM) models data as a mix of Gaussian distributions, assigning probabilities to each data point rather than fixed clusters. This flexibility makes it effective for overlapping or non-linear data structures. **Hierarchical Density-Based Spatial Clustering of Applications with Noise** (HDBSCAN) extends Density-Based

Spatial Clustering of Applications with Noise (DBSCAN) by detecting clusters of varying densities without requiring a predefined number. It handles irregular structures well and filters out noise by distinguishing core clusters from outliers according to Singgalen (2024). Main libraries used are Pandas (pandas development team (2020)), Scikit-learn (Pedregosa et al. (2011)), matplotlib (Hunter (2007)), and seaborn (Waskom (2021)) for data handling, analysis, its clustering, scaling and plotting.

4 Machine Learning Applications

4.1 K-means

Since clustering algorithms like K-Means are sensitive to scale according to Wongoutong (2024), the numerical features are standardized using StandardScaler from scikit-learn, Pedregosa et al. (2011), to avoid biased results. Variables used are **lead_time** (0 to 365 days), and **ADR** (price from 50 to 500).

To determine the optimal number of clusters, the Elbow method is used in Figure A8.

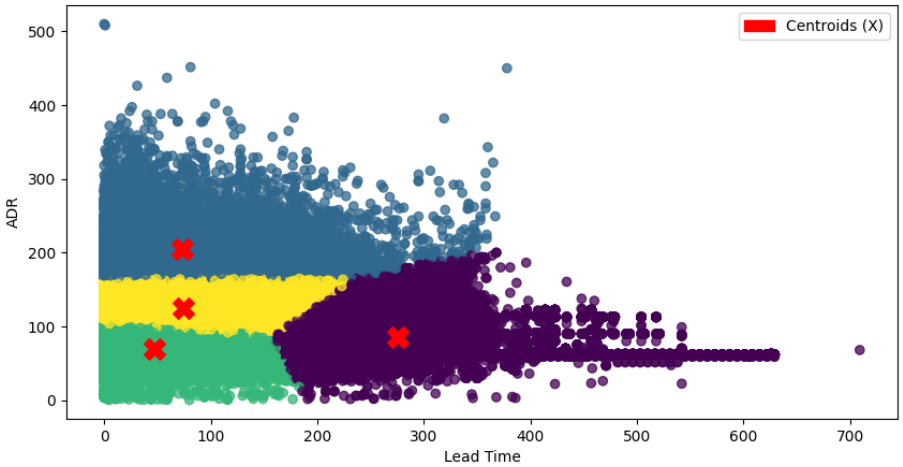

Figure 2: K-Means Clustering of Lead Time vs. ADR with Centroids.

As seen in Figure 2, K-Means clustering revealed clear customer segments.
Green: short lead time, low ADR (last-minute budget travelers or discounts).
Yellow: medium lead time, mid ADR (balanced planners).
Blue: short lead time, high ADR (premium travelers or urgent bookings).
Purple: long lead time, low ADR (early planners on budget).

4.2 GMM and HDBSCAN

To ensure a fair comparison between advanced clustering models and K-Means, we kept non-model-related actions, such as data preparation, as consistent with the K-Means

solution as possible. For dimensional consistency, we limited the dataset to the two key features (lead time and ADR). For GMM, we determined the optimal number of clusters using the knee method on the Bayesian Information Criterion (BIC) as seen in A9.

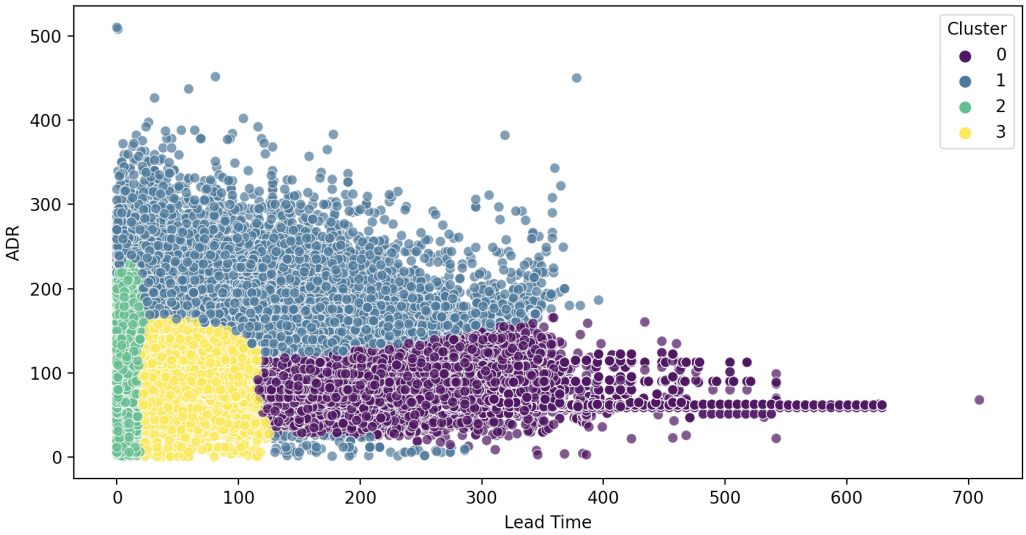

Figure 3: GMM Clustering of Lead Time vs. ADR

Besides largely aligning with K-Means, it showed that GMM suggests a more fluid transition between urgent (green) and mid-range (yellow) bookings, indicating that hotels could use dynamic pricing strategies to shift customers between these segments.

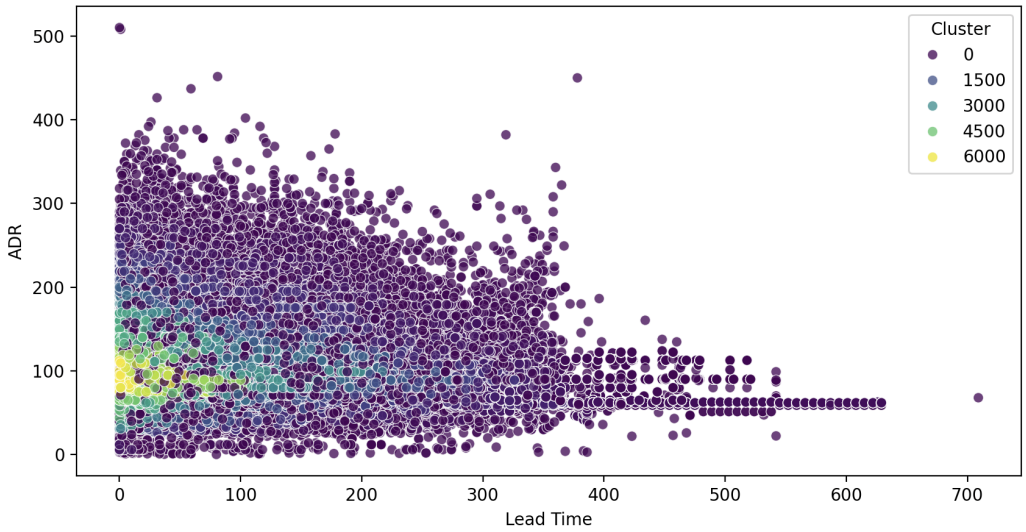

Figure 4: HDBSCAN Clustering of Lead Time vs. ADR.

HDBSCAN's clustering output was less intuitive. The yellow and green clusters in the lower-left suggested a pattern among last-minute, budget-conscious guests, while the purple clusters were more scattered, possibly indicating noise. Since the dataset was pre-processed to reduce noise, HDBSCAN's strength in handling noisy data was less observable.

4.3 Technical Comparison

Figure A10 compares all three models from a technical view. Log-likelihood measures how well the data fits the model, with GMM vastly outperforming K-Means. This metric does not apply to the probabilistic HDBSCAN. Silhouette score reflects cluster separation, with higher values meaning more distinct groupings. While K-Means and HDBSCAN performed reasonably well, GMM produced overlapping clusters. K-Means was the fastest. Both, K-Means and GMM formed well-structured clusters, suitable for datasets with a reasonable number of groups. In contrast, HDBSCAN fragmented the data into over 5,000 micro-clusters and classified 22% as noise (orange section), showing its fine-graining ability but challenging interpretation.

5 Conclusion and Shortcomings

This business carries high risk, with 1 in 3 bookings canceled. Hotels must analyze customer clusters to optimize revenue, with lead time and ADR as key indicators in this dataset. GMM and K-Means clustering reveal two key segments: short lead time, high ADR and long lead time, low ADR. To maximize revenue, hotels should use dynamic pricing, stricter policies, and personalized services for premium travelers, while applying flexible policies, overbooking, and upselling for budget guests. The revenue management strategy should also focus on covering volatile short-term bookings of high ADR customers, and targeted marketing in January and July, the peak canceled months.

Incorporating economic trends, competitor pricing, and major events could refine forecasting as researched in Antonio et al. (2019). While K-Means, GMM, and HDBSCAN were used, advanced models like XGBoost, or deep learning may better capture nonlinear patterns, improving cancellation predictions and revenue strategies.

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A Appendix

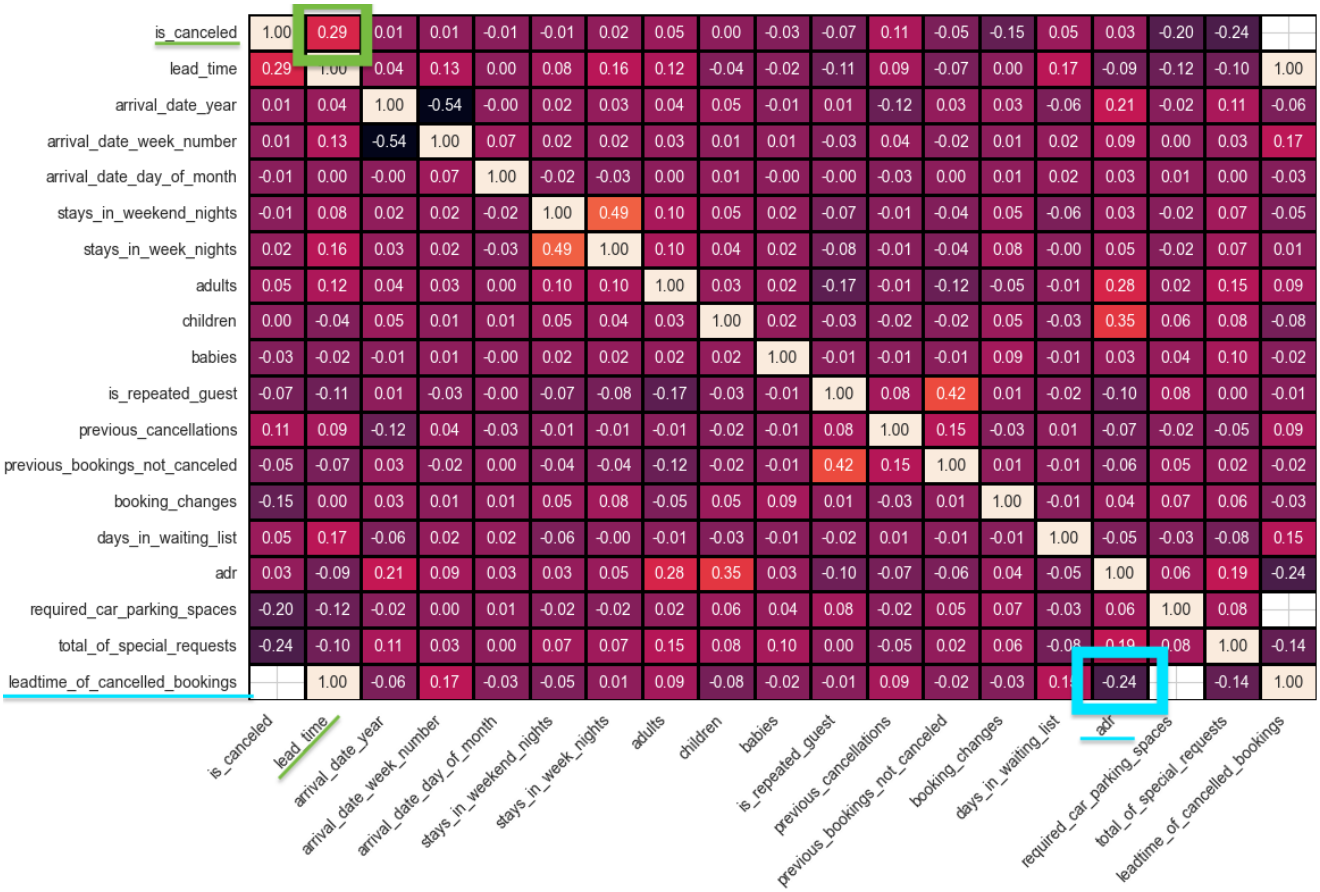


Figure A5: Correlation Heatmap Between Cancellation Behavior, Lead Time, and ADR.

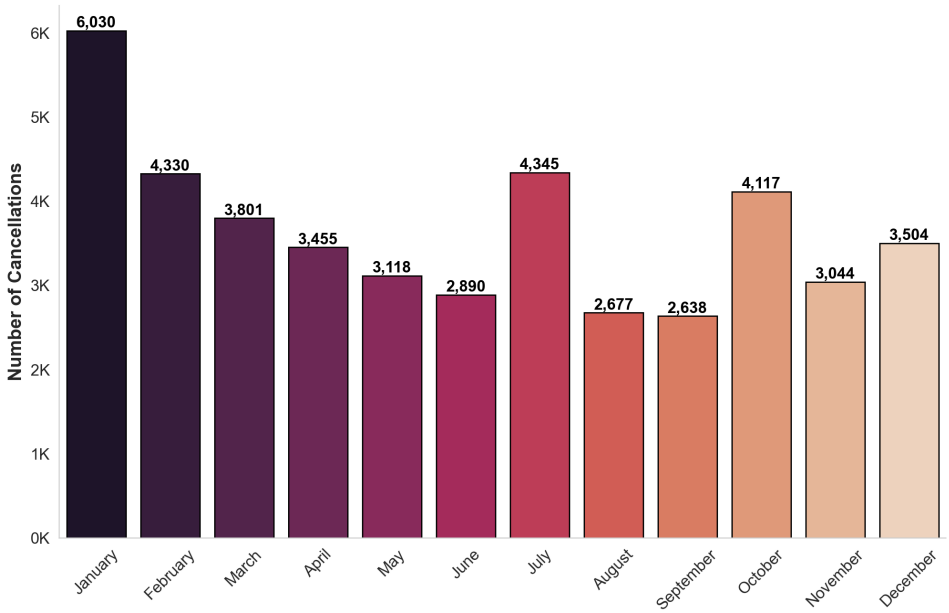


Figure A6: Monthly Cancellation: Numbers Represent Bookings Revoked in Each Month.

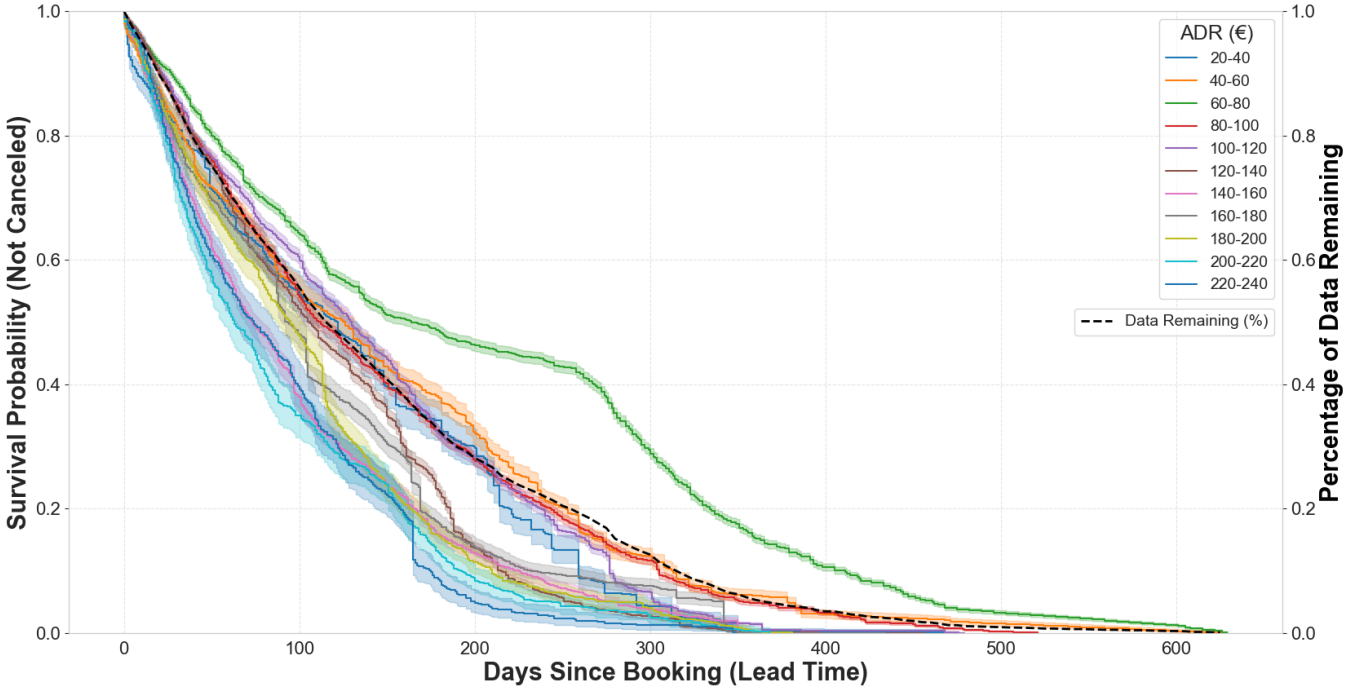


Figure A7: Survival Analysis of Bookings by Lead Time and ADR Ranges.

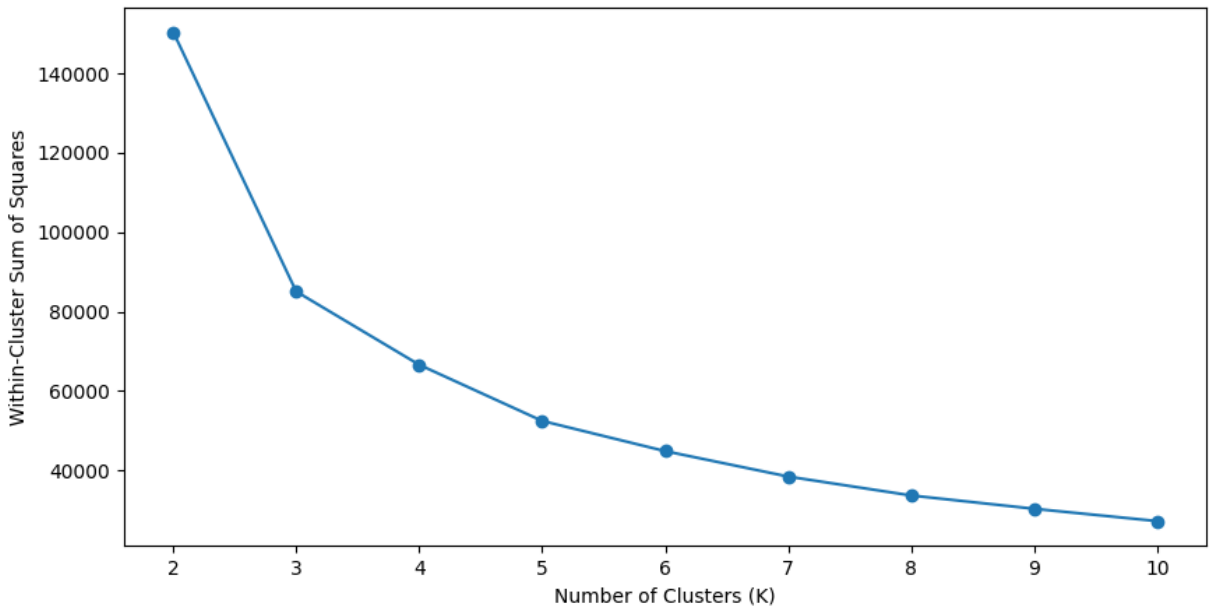


Figure A8: Elbow Method to Determine Optimal Number of Clusters for K-Means.

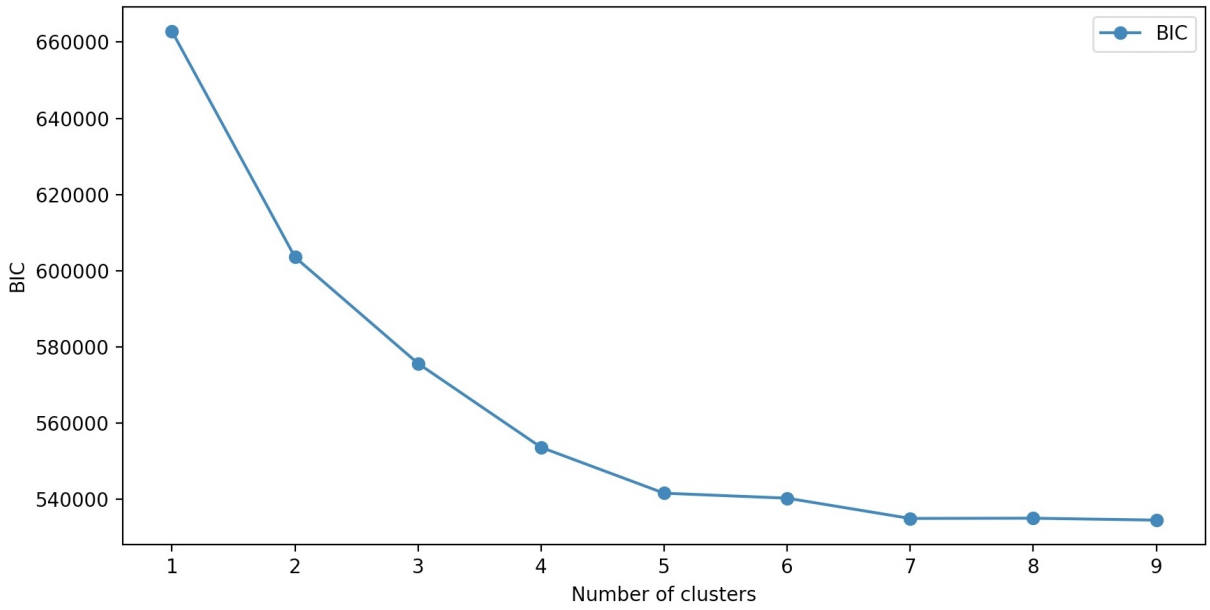


Figure A9: BIC to Determine Optimal Cluster Selection for GMM.

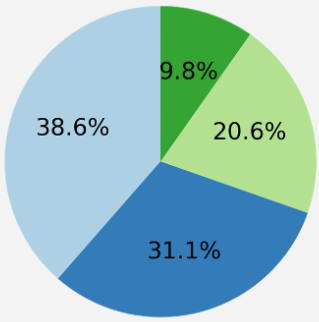
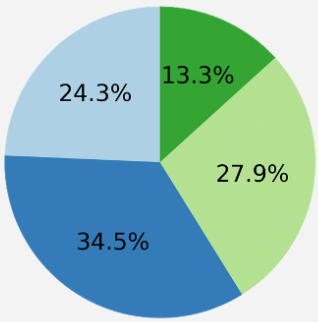
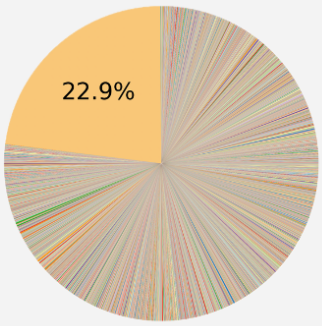
	KMeans	GMM	HDBSCAN
Fit Time (s)	0.3221	9.0561	2.9104
Log-Likelihood	-66603.91	-2.37	N/A
Silhouette Score	0.3842	0.1960	0.3406
Cluster Size Distribution			

Figure A10: Technical Comparative Performance of K-Means, GMM, and HDBSCAN Clustering.